MODULE 45 · DOMAIN 5 - OPERATIONAL EXCELLENCE AND AUTOMATION

Observability Foundations

Metrics, logs and traces from every workload and every account, gathered with open standards and kept for as long as you need at the lowest cost.

DOMAIN 5 of 5 EXAM WEIGHT 20% MODULE 45 of 52

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Instrument once with **OpenTelemetry**, view every account from **one place**, and keep each signal in the cheapest store that meets its query needs.

Three signals

SIGNAL	ANSWERS	AWS SERVICE
Metrics	How much, how fast, is it healthy?	CloudWatch metrics
Logs	What exactly happened?	CloudWatch Logs
Traces	Where did the time go across services?	X-Ray, Transaction Search

ON THE EXAM

Embedded metric format turns structured log lines into metrics without API calls.

X-Ray SDK or OpenTelemetry?

X-RAY SDKS AND DAEMON

Maintenance mode

- Maintenance since 25 Feb 2026
- End of support 25 Feb 2027
- Security fixes only
- Migrate existing code

ADOT AND OPENTELEMETRY

The supported path

- Vendor-neutral instrumentation
- Traces, metrics and logs
- Sends to X-Ray and CloudWatch
- Auto-instrumentation available

THE TIEBREAKER

New instrumentation: OpenTelemetry through ADOT, not the X-Ray SDK.

Container Insights

PLATFORM	HOW IT IS ENABLED	WHAT IT ADDS
EKS	CloudWatch Observability add-on	Pod, container and control plane metrics
ECS	Cluster setting, enhanced observability	Task and container metrics
Both	Curated dashboards	Drill down from cluster to container

ON THE EXAM

Per-container CPU, memory and network with little setup: Container Insights.

Seeing every account

CROSS-ACCOUNT OBSERVABILITY

View in place

- Monitoring account linked to sources
- Metrics, logs and traces
- No data copied
- Set up across an organisation

LOG SUBSCRIPTIONS

Copy and stream

- Subscription filter to a destination
- Kinesis, Firehose or Lambda
- Account-level filters cover every group
- Feeds a SIEM or archive

THE TIEBREAKER

Look across accounts: cross-account observability. Move the data somewhere: subscriptions.

Keeping logs at the right cost

STORE	QUERY SPEED	SUITS
CloudWatch Logs Standard	Seconds, full features	Active troubleshooting
CloudWatch Logs Infrequent Access	Seconds, fewer features	Occasional queries
S3 Standard-IA, Glacier Instant	Milliseconds, Athena	Monthly analysis
S3 Glacier Flexible Retrieval	Minutes to hours	Rare audits
S3 Glacier Deep Archive	Within 12 hours	Years of retention

ON THE EXAM

Retention in years, query within a day: Firehose to S3, lifecycle to Deep Archive.

Reading the scenario

WHEN THE QUESTION SAYS	IT IS TESTING
New tracing, vendor-neutral	ADOT and OpenTelemetry
One view of all accounts, no copying	Cross-account observability
Stream logs to a security account	Subscription filters
Per-pod or per-task metrics	Container Insights
Keep for seven years, rarely read	S3 lifecycle to Glacier
Metrics from Lambda without API calls	Embedded metric format

ON THE EXAM

How it is collected, where it is viewed, where it is kept.

> Corvane Retail is adding distributed tracing to 40 services on ECS and Lambda. The lead developer proposes the X-Ray SDK and daemon, which an older project used.
>
> The company wants instrumentation it will not need to replace, and may later send traces to a second tool.

What should the solutions architect recommend?

A. The X-Ray SDK and daemon, as proposed

B. VPC Flow Logs with Athena

C. CloudTrail data events

D. OpenTelemetry instrumentation through ADOT, sending traces to X-Ray and CloudWatch

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A X-Ray SDK

In maintenance mode; end of support February 2027.

B Flow Logs

Network records, not request traces.

C CloudTrail

API audit, not application traces.

D ADOT

Supported, vendor-neutral, exports to several tools.

THE SIGNAL

New tracing that will last and stay portable: OpenTelemetry through ADOT.

> Corvane Retail has 60 workload accounts. During incidents, operators switch roles between accounts to read dashboards, logs and traces.
>
> The operations team wants one place to search and correlate all three signals across accounts, without copying the data.

What should the solutions architect implement?

A. CloudWatch cross-account observability, with a central monitoring account linked to every workload account

B. Subscription filters sending all logs to a central account

C. A dashboard in each account

D. Amazon OpenSearch Service with every log copied in

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A **Cross-account observability**
Metrics, logs and traces viewed in place.

B **Subscriptions**
Copies logs only; no metrics or traces.

C **Per-account dashboards**
Still switching accounts.

D **OpenSearch**
Copies data and adds a cluster to run.

THE SIGNAL

All signals, all accounts, one view, no copying: cross-account observability.

> Corvane Retail's security team runs a SIEM in a dedicated security account. It needs every application log group in 60 accounts streamed to it in near real time.
>
> New log groups must be included automatically.

What should the solutions architect configure?

A. Cross-account observability linking every account

B. Account-level subscription filters in each account, sending to a cross-account destination backed by Kinesis Data Streams in the security account

C. A daily export of log groups to S3

D. CloudTrail organisation trail

◆ PAUSE AND DECIDE

What should the solutions architect configure?

A Cross-account observability

Views data; does not deliver it to a SIEM.

B Account-level subscriptions

Streams every group, new ones included.

C Daily export

Not near real time.

D CloudTrail

API events, not application logs.

THE SIGNAL

Deliver logs to another account's tool, continuously: subscription filters.

> Corvane Retail's EKS clusters show healthy node metrics, but users report slow checkouts. Engineers cannot see which pods are short of memory or being throttled.
>
> They want pod and container metrics with ready-made dashboards, with as little setup as possible.

What should the solutions architect do?

A Publish custom metrics from each application with PutMetricData

B Enable detailed monitoring on the worker nodes

C Install the CloudWatch Observability add-on to enable Container Insights with enhanced observability

D Turn on VPC Flow Logs

◆ PAUSE AND DECIDE

What should the solutions architect do?

A **Custom metrics**
Code changes in every service; no dashboards.

B **Detailed monitoring**
More frequent node metrics, still no pod view.

C **Container Insights**
Pod and container metrics with dashboards, one add-on.

D **Flow Logs**
Network records, not resource use.

THE SIGNAL

Pod or container resource metrics, little setup: Container Insights.

> Corvane Retail must keep payment application logs for seven years. After 90 days they are read only for audits, about once a year, and auditors accept results within a day.
>
> The company wants the lowest storage cost.

What should the solutions architect design?

A. Stream logs through Firehose to S3, with a lifecycle rule moving objects to S3 Glacier Deep Archive after 90 days

B. Set CloudWatch Logs retention to seven years

C. Use the CloudWatch Logs Infrequent Access class for seven years

D. Store logs in S3 Glacier Instant Retrieval

◆ PAUSE AND DECIDE

What should the solutions architect design?

A S3 to Deep Archive

Lowest cost; retrieval within 12 hours meets the day.

B CloudWatch Standard

Most expensive way to store cold logs.

C Logs Infrequent Access

Cheaper ingestion, still far above Deep Archive storage.

D Glacier Instant

Pays for millisecond access nobody needs.

THE SIGNAL

Years of retention, read within a day: S3 lifecycle to Glacier Deep Archive.

> Corvane Retail's checkout Lambda function needs custom metrics for basket value and payment provider, broken down by several dimensions.
>
> Calling the CloudWatch API on every invocation added latency and throttling errors.

What should the solutions architect recommend?

A Batch PutMetricData calls in a background thread

B X-Ray annotations

C Write the values to DynamoDB and graph them later

D Write the metrics as structured log lines in CloudWatch embedded metric format

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Background batching

Lambda may freeze before the thread sends.

B X-Ray annotations

Index traces; they are not metrics.

C DynamoDB

Builds a metrics system by hand.

D Embedded metric format

Logs written asynchronously become metrics.

THE SIGNAL

Custom metrics from Lambda without API calls: embedded metric format.

What to carry out of this module

- **ADOT and OpenTelemetry** for new instrumentation; X-Ray SDK ends Feb 2027.
- **Cross-account observability** views; **subscriptions** copy.
- **Container Insights** for pod and task metrics.
- **Firehose to S3**, lifecycle to Glacier, for years of logs.
- **Embedded metric format** for metrics without API calls.

> ON THE EXAM
>
> How it is collected, where it is viewed, where it is kept.